# Multi-Fidelity Gaussian Process for Time-Series with application to industrial processes

Ugo LABBÉ [1,2,3], Josselin GARNIER [2], Mickaël BINOIS [3], Amina CHORFI [1] & Mayra HERNANDEZ [1]

[1]Manufacture de Pneu Michelin, 63000 Clermont-Ferrand, France [2]Centre de Mathématiques Appliquées, Ecole polytechnique, Institut Polytechnique de Paris, 91120 Palaiseau, France [3]Université Cote d'Azur, Inria, CNRS, LJAD, 06902 Sophia Antipolis, France

## Context

- **Accurate forecasting** and **uncertainty quantification** of **time series**, of different time length, are of extreme importance for process control and safety in industrial processes.
- **High-Fidelity (HF) measurements** are often scarce due to acquisition costs or availability, while **Low-Fidelity (LF) simulations** is abundant but imperfect.
- **Multi-Fidelity Gaussian Process (MFGP)** frameworks can leverage this low-cost data to improve prediction and provide accurate uncertainty quantification [2, 4].

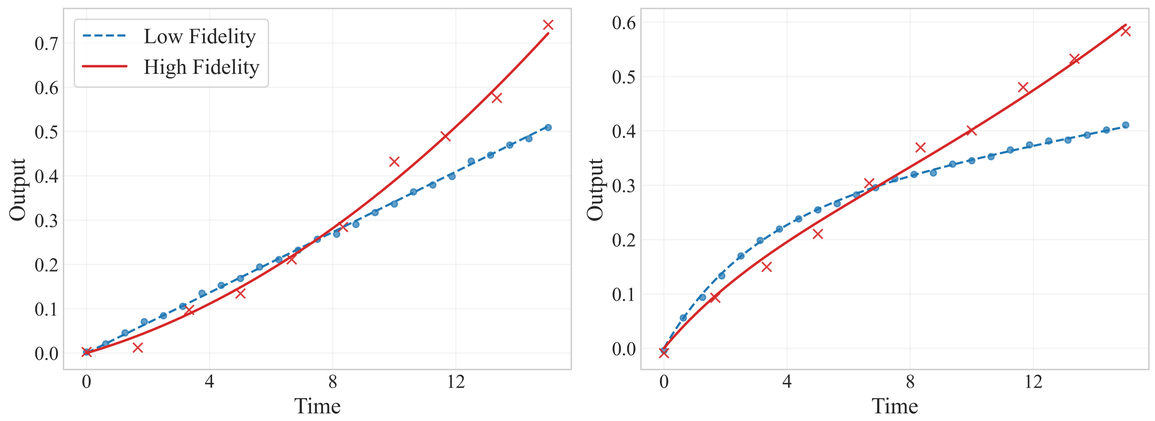


Figure 1. Samples of cases 1 and 2, with LF functions in blue and HF functions in red.

## Multi-Fidelity Gaussian Process

Commonly used **Multi-Fidelity Gaussian Process** formulation assumes a **recursive autoregressive relationship** [1, 5] of the form:

$$f_{\text{HF}}(\mathbf{x}) = \rho(\mathbf{x}) \cdot f_{\text{LF}}(\mathbf{x}) + \delta(\mathbf{x}) \tag{1}$$

With:

- $\rho(\mathbf{x}) = \mathbf{g}_{\text{LF}(\mathbf{x})}{}^{\top}\beta_\rho$, a scaling factor
- $f_{\text{LF}(\mathbf{x})}$ Gaussian process modeling the LF simulations with covariance structure $k_{\text{LF}}$, parameters are estimated using maximum likelihood estimation.
- $\delta(\mathbf{x})$, the discrepancy function modeled as a GP with mean $m_\delta(\mathbf{x}) = \mathbf{f}_\delta(\mathbf{x})^\top\beta_\delta$ and covariance $k_\delta(\mathbf{x}, \mathbf{x}')$, parameters are optimized via the expectation-maximization (EM) algorithm [1].
- $f_{\text{LF}}(\mathbf{x}) \perp \delta(\mathbf{x})$, meaning they are independent
- HF data is not necessarily nested within LF data thanks to the use of EM.

However, this formulation inherently assumes a **static, linear correlation between fidelities**. It **often proves insufficient for dynamic industrial systems**, where the discrepancy between simulation and physical reality is non-stationary and evolves over time.

## Time-Aware Multi-Fidelity Gaussian Process

To overcome these limitations, we propose a **Time-Aware Multi-Fidelity Gaussian Process** framework:

- **Decomposition of the input space** into a temporal dimension and a state-variable dimension, $\mathbf{x} = (t, \mathbf{z})$.
  - $t$ represents the temporal dimension.
  - $\mathbf{z}$ represents the physical state variables.
  - For each sampled state $\mathbf{z}^{(i)}$, the full output trajectory is observed at all time points $t_1, \dots, t_{N_t}$.
- **Two proposed model variants** with time-evolving coupling, differing in their dependence on $\mathbf{z}$:

$$\rho(t) = \mathbf{g}_{\text{LF}}(t)^\top\beta_\rho \qquad \rho(t, \mathbf{z}) = \mathbf{g}_{\text{LF}}(t, t \cdot \mathbf{z})^\top\beta_\rho \tag{2}$$

$$m_\delta(t) = \mathbf{f}_\delta(t)^\top\beta_\delta \qquad m_\delta(t, \mathbf{z}) = \mathbf{f}_\delta(t, t \cdot \mathbf{z})^\top\beta_\delta \tag{3}$$

- **Reduction of computational costs** of training the Low-Fidelity data, using a Kronecker product structure for the Gaussian process covariance matrix [8].

$$k_{\text{LF}} = k_{state_{\text{LF}}} \otimes k_{time_{\text{LF}}} \tag{4}$$

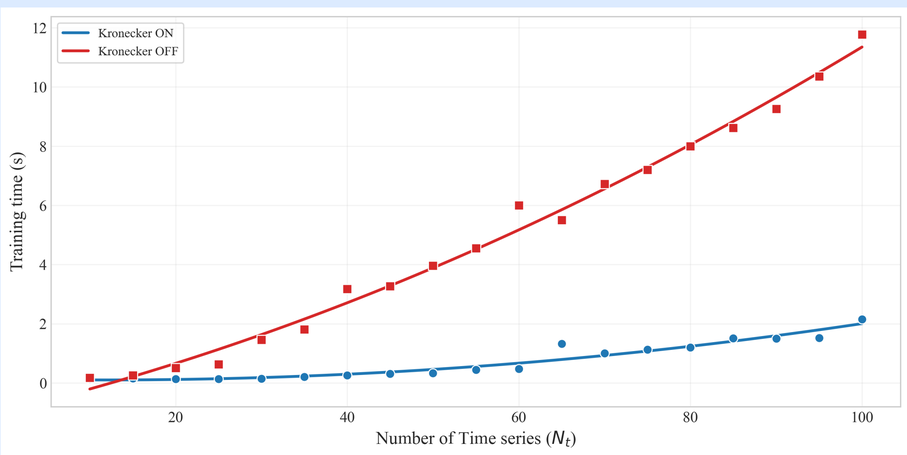


Figure 2. Training time for the Low-Fidelity GP with and without the use of a Kronecker product structure.

## Benchmark

To assess the performance of our Time-Aware MFGP models, we developped different cases (see Figure 1) of different dimensions, representing credible physical systems with non-linear and time-evolving relationships between fidelities:

- 30 Low-Fidelity time-series.
- Varying amount of High-Fidelity time-series.
- Respectively 1% and 5% of the Low-Fidelity and High-Fidelity signal standard deviation as noise.

We compare the performance of our models to a standard MFGP model, as well as to a HF-only GP baseline, in terms of **predictive accuracy** ($RMSE$) and **uncertainty quantification** ($IAE_\alpha$) [7].

$$\text{RMSE} = \left\{ \frac{1}{n} \sum_{i=1}^{n} \left( y(\mathbf{x}^{(i)}) - \hat{y}_{-i}(\mathbf{x}^{(i)}) \right)^2 \right\}^{1/2} \tag{5}$$

$$IAE_\alpha = \int_0^1 |\widehat{\Delta}(\alpha) - \alpha| \, d\alpha. \tag{6}$$

With:

- $\widehat{\Delta}(\alpha) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{y_i \in PI_{\alpha,-i}(\mathbf{x}^{(i)})\}$, the empirical coverage of the leave-one-out prediction interval at confidence level $\alpha$

## Results

Metrics are computed on a test set of 30 High-Fidelity noiseless time-series, with 30 repetitions for each case and budget.

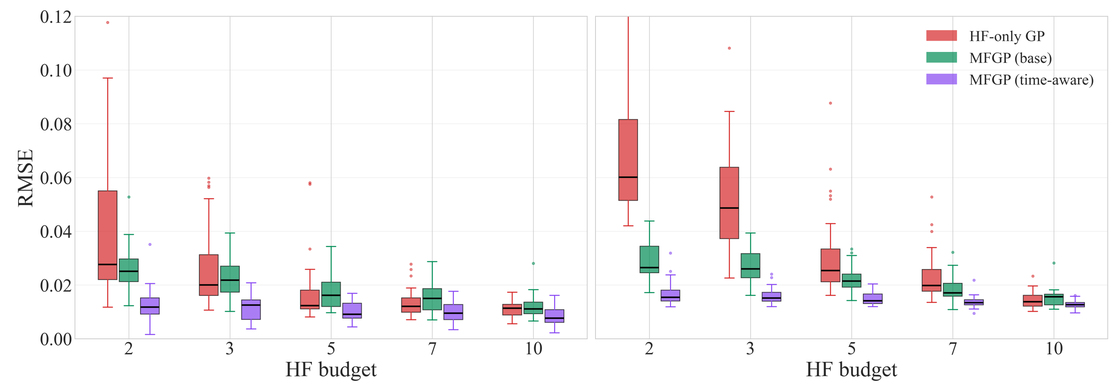


Figure 3. RMSE comparison of the different models for cases 1 and 2.

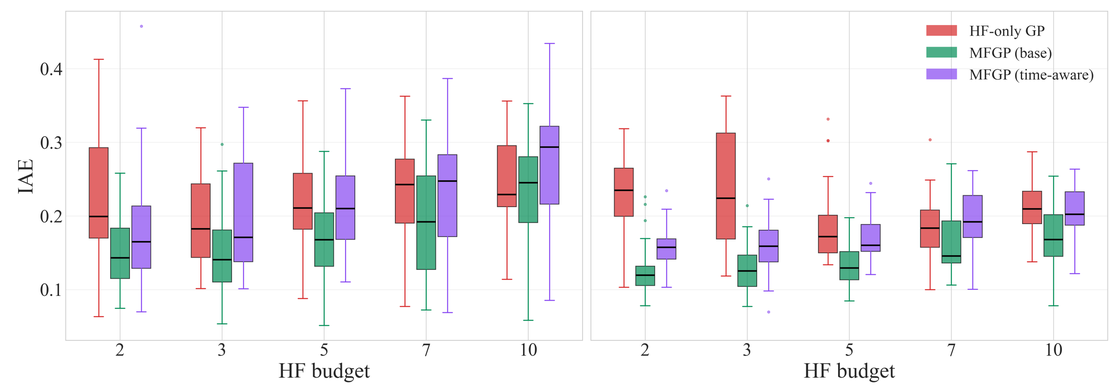


Figure 4. IAE comparison of the different models for cases 1 and 2.

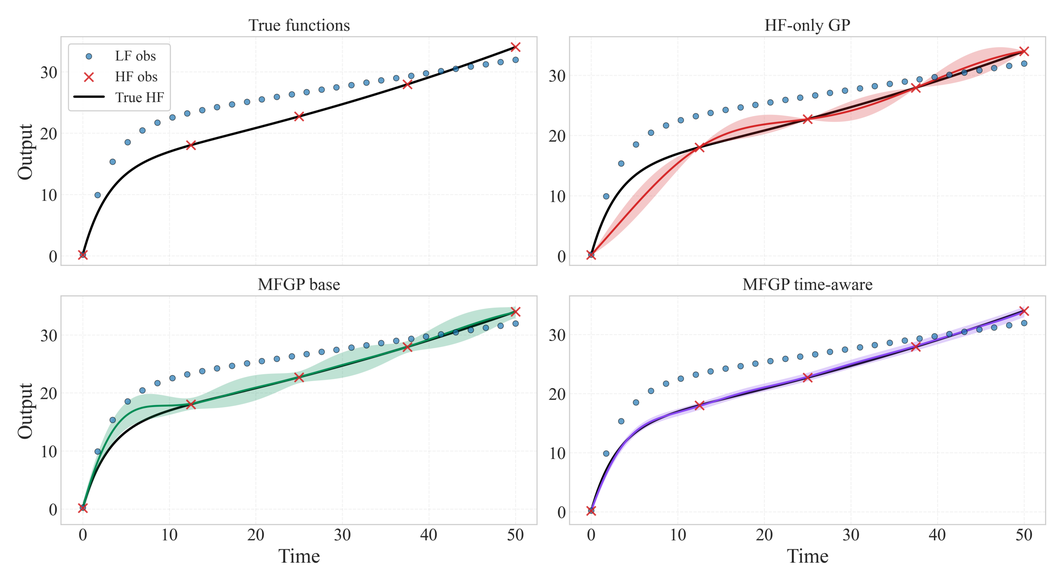


Figure 5. Example of predictions of the HF-only GP and MFGP base model against a Time-Aware MFGP model

## Preliminary conclusions

- **Time-aware MFGP models outperform standard MFGP and HF-only baselines** in terms of predictive accuracy, even after sufficient High-Fidelity training data. Uncertainty quantification performance is more mixed.
- The use of a **Kronecker product structure for the covariance matrix significantly reduces training time** without compromising predictive performance, making it a viable option for large-scale time-series data.

## Future Steps

- Further work involves using **more flexible covariance structures to better predict time-series** [3, 10].
- **Constraining the time-series with physical constraints**, in the context of Multi-Fidelity is a promising area of research and has to be further explored [6, 9].

## References

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ugolabbe.github.io

ugo.labbe@michelin.com